

Yikai Tang

🏠 [Homepage](#) | ✉ wintermeloooo@gmail.com | 🌐 [WinterMeloooo](#)

Education

Carnegie Mellon University (CMU)

Aug 2026 – Present

Master of Science in Robotics

Shanghai Jiao Tong University (SJTU)

Sep 2022 – Jun 2026

Advisor: **Prof. Yunbo Wang**

B.E. in Computer Science and Technology (IEEE Honor Class)

- **GPA: 3.8**
- **27 A/A+** Courses: Design and Analysis of Algorithms, Operating System, Discrete Mathematics, etc.
- GRE: V160, Q168; TOEFL: 110

University of California, Berkeley

Jan 2025 – May 2025

Advisor: **Prof. Jitendra Malik**

Visiting Student, BGA Program

- CS 184: Foundations of Computer Graphics — **A**
- CS 61A: Structure and Interpretation of Computer Programs — **A**
- EECS 126: Probability and Random Processes — **A-**

Publications

AXIS

Axis Robotics | 2026

A Growable Community-Driven Data Engine for Scalable Robot Manipulation

M. Zhao*, D. Huang*, **Y. Tang***, P. Li*, M. Yan*, R. Zhuang*, Y. Huang*, J. Wang, H. Zhai, T. Zhou, R. Zhang, Z. Luo, Y. Huang, J. Yang, J. Li

- Built a **growable, community-driven data engine** for browser-based robot demonstration collection, automated task generation, validation, refinement, and augmentation.
- Released a benchmark with **207 tasks and 50K+ trajectories**; AXIS continual pretraining improved $\pi_{0.5}$ robustness on LIBERO-Plus.

DIPOLE

BAIR, UC Berkeley | Mar 2025 – Sep 2025

Fusing Vision and Geometry for Robust Visuomotor Generalization

Y. Tang*, H. Geng*, J. Jia, Y. Hu, S. Zang, J. Yang, P. Abbeel, J. Malik

- Proposed a **modality-aware dropout mechanism** enabling robust RGB–point-cloud fusion within diffusion policies.
- Built a unified visual–geometric latent space that improves generalization across lighting, materials, viewpoints, and object configurations.
- Demonstrated strong, robust, state-of-the-art performance through geometry-informed policy learning.

ISF-VLA

IROS 2026 | BAIR, UC Berkeley | Sep 2025 – Nov 2025

An Implicit 3D Vision-Language-Action Model Capitalizing on 3D-Aware Vision Foundation Models through Implicit Spatial Fusion

S. Zang*, S. Wei*, H. Geng, **Y. Tang**, B. An, S. Jayavelu, P. Abbeel, J. Malik

- Introduced **implicit 3D spatial fusion** to integrate geometric cues into RGB–point-cloud fusion **for VLA**.
- Achieved significant gains in cross-scene and cross-viewpoint robustness through 3D-grounded multi-modal representation learning.
- Enabled stable and expressive fusion for large pretrained visual–language backbones in manipulation.

RoST3R

BAIR, UC Berkeley | Mar 2025 – Sep 2025

Robot-Aware Dynamic 3D Reconstruction from RGB Input for Robotic Manipulation

Y. Han, H. Geng, J. Zhang, Z. Chen, **Y. Tang**, C. Cheng, P. Li, T. Darrell, P. Abbeel, J. Malik, Y. Wang

- Developed a dynamic, metric-scale 3D reconstruction pipeline from monocular RGB tailored for robotic manipulation.
- Enabled consistent 3D scene understanding with temporally stable geometry beneficial for downstream control.
- Delivered robot-aware reconstruction that integrates motion cues for improved dynamic-scene fidelity.

Experience

Axis Robotics

Feb 2026 – Present

Position: AI Research & Engineering Lead

AXIS* — Growable community-driven data engine for scalable robot manipulation

BAIR, UC Berkeley

Mar 2025 – Nov 2025

Position: Research Assistant

Advisor: **Prof. Jitendra Malik**

DIPOLE* — Multimodal diffusion policy with modality-aware fusion

Mar 2025 – Sep 2025

RoST3R — Robot-aware dynamic 3D reconstruction from RGB

Mar 2025 – Sep 2025

ISF-VLA — Implicit 3D spatial fusion for VLA models

Sep 2025 – Nov 2025

Yunbo Lab, SJTU

Oct 2023 – Dec 2024

Position: Research Assistant

Advisor: **Prof. Yunbo Wang**

DeformSDF* — Physics-aware SDF reconstruction for highly deformable objects

Aug 2024 – Dec 2024

Intuitive Physics Fluids — Multi-view real-world fluid reconstruction

Oct 2023 – Jul 2024

*First / Co-First Author

Projects

Game Design with olcPixelGameEngine

Jun 2023 – Jul 2023

Advisor: Yuting Wang

Tech: C++

- Designed and implemented an SLG game with custom UI and unit systems.
- Implemented pixel-level visual effects for movement, attacks, and explosions.

News Crawler and Search Engine

Dec 2023 – Jan 2024

Advisor: Ya Zhang

Tech: Flask, BeautifulSoup, Lucene, Face_Recognition

- Built a multithreaded crawler collecting 10k+ news articles and images.
- Implemented a Lucene-based search backend with a Flask web interface.

Structured Macro Support for WhileDB

Dec 2024 – Jan 2025

Advisor: Qinxiang Cao

Tech: Flex, Bison, C#

- Extended WhileDB with structured macros and function calls while preserving AST integrity.

Honors and Awards

SJTU Merit Scholarship for Academic Excellence (Category C) — 2023

Class Student Leader in Academics — 2022

Professional Service

Workshop Organizer: **[NeurIPS 2026]** Embodied World Models for Decision Making 

Leadership and Activities

Debate Team, SJTU — Competed in 30+ tournaments; mentored 20+ members.

Technical Skills

Python, C++, Coq; PyTorch, ROS2, Transformers, Open3D; Linux, Git